

WASP-2b

R-1022

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_151201 · created 2026-07-30T04:48:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457357.5975 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.046
Transit depth (Rp/Rs)^2	2.19 +/- 1.38 [%]
Orbital Inclination (inc)	84.09 +/- 2.84
Airmass coefficient 1 (a1)	0.28 +/- 0.15
Airmass coefficient 2 (a2)	0.48 +/- 0.14
Scatter in the residuals of the lightcurve fit is	52.15 %
Best Comparison Star	#2 - [179, 290]
Optimal Aperture	1.73
Optimal Annulus	6.79
Transit Duration (day)	0.047 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.046 vs 0.1326 (deviation 0.27σ)
Duration fit vs published	0.047 d vs 0.0737 d (deviation 0.71σ)
Mid-time O–C	10.0 min
Depth SNR	2.6
Residual scatter	52.15 %

Light curve

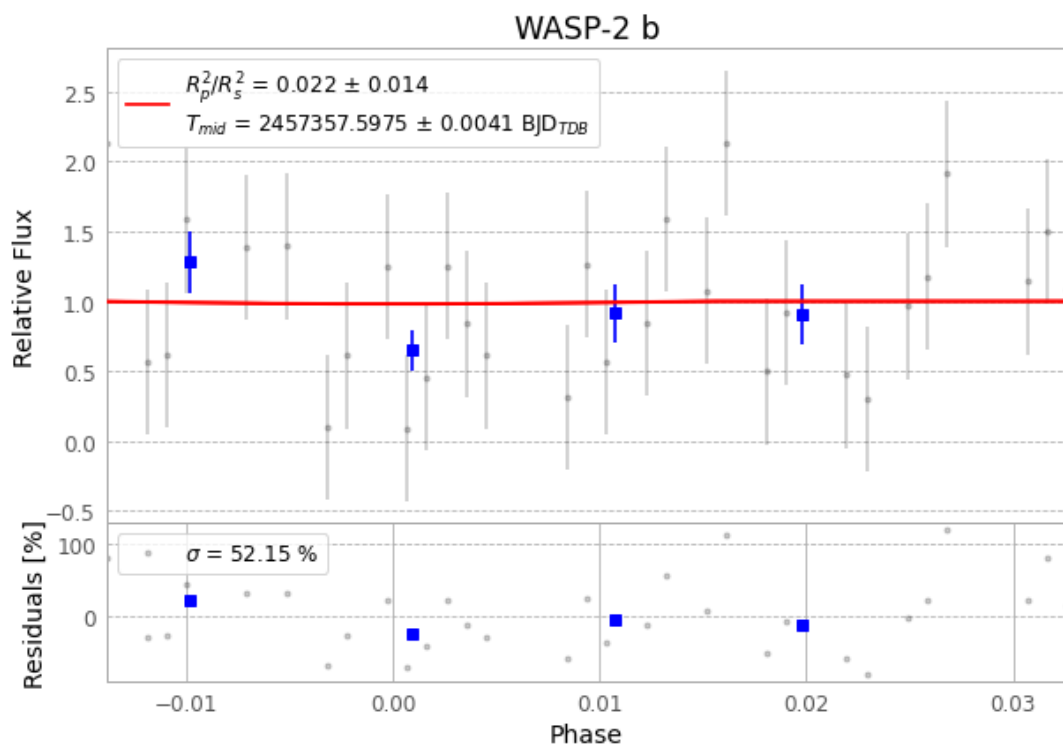


Figure 1 — FinalLightCurve_WASP-2 b_2015-11-30.png (SHA-256 7f4d5ee6a8405295...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 429], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f1a7583f7a2c1321...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 902034c

Data provenance

49 FITS frames (32 MB), WASP-2151201013912.FITS → WASP-2151201040312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-151201.log (146bb92d8c15...), FinalLightCurve_WASP-2 b_2015-11-30.png (7f4d5ee6a840...), FinalLightCurve_WASP-2 b_2015-11-30.csv (365ed1122453...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 04:48Z.